

Chapter 7

Periodic properties of the elements

1. Electrons in the 1s subshell are much closer to the nucleus in Ar than in He due to the larger A in Ar.

- A) nuclear charge
- B) paramagnetism
- C) diamagnetism
- D) Hund's rule
- E) azimuthal quantum number

2. Screening of the nuclear charge by core electrons in atoms is B.

- A) less efficient than that by valence electrons
- B) more efficient than that by valence electrons
- C) essentially identical to that by valence electrons
- D) responsible for a general decrease in atomic radius going down a group
- E) both essentially identical to that by valence electrons and responsible for a general decrease in atomic radius going down a group

A: electrons for which n is larger than the value of n for the electron of interest contribute 0 to the value of S .

B: electrons with the same value of n as the electron of interest contribute 0.35 to the value of S , except for the $[1s]$ group, where the other electron contributes only 0.30.

C: electrons that have principle quantum number $n-1$ contribute 0.85 to the value of S .

D: electrons that have even smaller principle quantum number contribute 1 to the value of S .

3. The atomic radius of main-group elements generally increases down a group because D.

- A) effective nuclear charge increases down a group
- B) effective nuclear charge decreases down a group
- C) effective nuclear charge zigzags down a group
- D) the principal quantum number of the valence orbitals increases
- E) both effective nuclear charge increases down a group and the principal quantum number of the valence orbitals increases

4. Of the following, which gives the correct order for atomic radius for Mg, Na, P, Si and Ar D ?

A) $\text{Mg} > \text{Na} > \text{P} > \text{Si} > \text{Ar}$

B) $\text{Ar} > \text{Si} > \text{P} > \text{Na} > \text{Mg}$

C) $\text{Si} > \text{P} > \text{Ar} > \text{Na} > \text{Mg}$

D) $\text{Na} > \text{Mg} > \text{Si} > \text{P} > \text{Ar}$

E) $\text{Ar} > \text{P} > \text{Si} > \text{Mg} > \text{Na}$

5. Which isoelectronic series is correctly arranged in order of increasing radius D?

A) $\text{K}^+ < \text{Ca}^{2+} < \text{Ar} < \text{Cl}^-$

B) $\text{Cl}^- < \text{Ar} < \text{K}^+ < \text{Ca}^{2+}$

C) $\text{Ca}^{2+} < \text{Ar} < \text{K}^+ < \text{Cl}^-$

D) $\text{Ca}^{2+} < \text{K}^+ < \text{Ar} < \text{Cl}^-$

E) $\text{Ca}^{2+} < \text{K}^+ < \text{Cl}^- < \text{Ar}$

6. Of the choices below, which gives the order for first ionization energies B?

A) $\text{Cl} > \text{S} > \text{Al} > \text{Ar} > \text{Si}$

B) $\text{Ar} > \text{Cl} > \text{S} > \text{Si} > \text{Al}$

C) $\text{Al} > \text{Si} > \text{S} > \text{Cl} > \text{Ar}$

D) $\text{Cl} > \text{S} > \text{Al} > \text{Si} > \text{Ar}$

E) $\text{S} > \text{Si} > \text{Cl} > \text{Al} > \text{Ar}$

- **Ionization energy:** the minimum energy required to remove an electron from the ground state of the isolated gaseous atom or ion; the higher the ionization energy, the more difficult it is to remove an electron.

7. A have the lowest first ionization energies of the groups listed.

A) Alkali metals

B) Transition elements

C) Halogens

D) Alkaline earth metals

E) Noble gases

8. The electron configuration belonging to the atom with the highest second ionization energy is A.

(i) $1s^2 2s^2 2p^6 3s^1$

(ii) $1s^2 2s^2 2p^6 3s^2$

(iii) $1s^2 2s^2 2p^6 3s^2 3p^1$

(iv) $1s^2$

(v) $1s^2$

A) (i)

B) (ii)

C) (iii)

D) (iv)

E) (v)

Table 7.2 Successive Values of Ionization Energies, I , for the Elements Sodium through Argon (kJ/mol)

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	496	4562	(inner-shell electrons)				
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

9. The electron configuration of the atom with the most negative electron affinity is E.

- (i) $1s^2 2s^2 2p^6 3s^1$
- (ii) $1s^2 2s^2 2p^6 3s^2$
- (iii) $1s^2 2s^2 2p^6 3s^2 3p^1$
- (iv) $1s^2 2s^2 2p^6 3s^2 3p^4$
- (v) $1s^2 2s^2 2p^6 3s^2 3p^5$

- A) (i)
- B) (ii)
- C) (iii)
- D) (iv)
- E) (v)

be absorbed to remove an electron.

Most atoms can also gain electrons to form anions. The energy change that occurs when an electron is added to a gaseous atom is called the **electron affinity** because it measures the attraction, or *affinity*, of the atom for the added electron. For most atoms, energy is *released* when an electron is added. For example, the addition of an electron

10. Chlorine is much more apt to exist as an anion than is sodium. This is because .

- A) chlorine is bigger than sodium
- B) chlorine has a greater ionization energy than sodium does
- C) chlorine has a greater electron affinity than sodium does
- D) chlorine is a gas and sodium is a solid
- E) chlorine is more metallic than sodium

11. In the generation of most anions, the energy change (kJ/mol) that _____ an electron is _____D_____.

- A) removes, positive
- B) adds, positive
- C) removes, negative
- D) adds, negative
- E) None of the above is correct

12. The list that correctly indicates the order of metallic character is
 C .

A) $B > N > C$

B) $F > Cl > S$

C) $Si > P > S$

D) $P > S > Se$

E) $Na > K > Rb$

13. Transition metals within a period differ mainly in the number of C electrons.

A) s

B) p

C) d

D) f

E) all of the above

14. Consider the general valence electron configuration of ns^2np^5 and the following statements:

- (i) Elements with this electron configuration are expected to form -1 anions.
- (ii) Elements with this electron configuration are expected to have large positive electron affinities.
- (iii) Elements with this electron configuration are nonmetals.
- (iv) Elements with this electron configuration form acidic oxides.

Which statements are true _____?

- A) (i) and (ii)
- B) (i), (ii), and (iii)
- C) (ii) and (iii)
- D) (i), (iii,) and (iv)
- E) All statements are true

15. This element is more reactive than lithium and magnesium but less reactive than potassium. This element is A .

- A) Na
- B) Rb
- C) Mn
- D) Be
- E) Fr

16. All of the halogens E.

- A) exist under ambient conditions as diatomic gases
- B) tend to form positive ions of several different charges
- C) tend to form negative ions of several different charges
- D) exhibit metallic character
- E) form salts with alkali metals with the formula MX